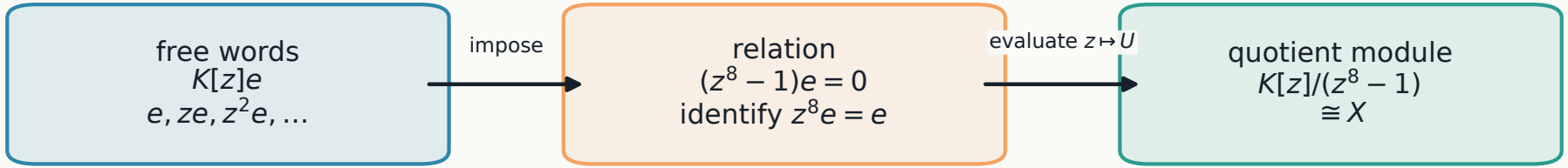
The orbit relation turns a free word algebra into a finite cyclic module



The finite relation is not an approximation. It is the exact kernel relation of the shift representation.

$U^8 = I$ **gives a complete normal form** $a_0 I + a_1 U + \cdots + a_7 U^7$**.**